

# Sepehr Mahmoudian

## SENIOR AI ENGINEER

Machine Learning • Knowledge Graphs • Multimodal & Embodied AI

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LINKEDIN [linkedin.com/in/sepehr-mahmoudian](https://linkedin.com/in/sepehr-mahmoudian)

Berlin-Brandenburg • available from Jan 2027 • German citizen • EN fluent / DE good / Persian native



### PROFILE

Builds custom agent harnesses for domain-specific applications, knowledge graphs and Graph RAG; evaluates LLMs, VLMs, VLAs and world models with rigorous, evidence-first engineering.

LLMs / VLMs • Knowledge Graphs • Agentic Vision • Multimodal AI • Scene Reconstruction  
CONTEXT ENGINEERING • DATA PROVENANCE • MODEL EVALUATION • REINFORCEMENT LEARNING

### AI ENGINEERING

10+ YEARS

RESEARCH TO PRODUCTION

PYTHON • RUST • AI / ML

## PROFESSIONAL EXPERIENCE

### Independent AI/ML Engineer & PhD Candidate

2020.04 - Present

Self-employed - PhD candidate at Technische Universität Darmstadt

Berlin & Darmstadt, Germany

- Building biophysical/functional neural models, PID (Partial Information Decomposition), Vision-Language-Action systems, VLA evaluation and neural control
- Building custom AI agent harnesses for domain-specific applications; multimodal 3D perception and auditable visualization

### Senior AI Engineer

2026.01 - 2026.03

Neural Frames UG

Berlin & Hamburg, Germany

- Evaluated LLMs, VLMs and end-to-end pipelines, tracing runs and results with Langfuse
- Contributed to Python pipelines for model evaluation and inference

### AI Engineering Resident

2025.09 - 2026.01

Merantix AI Campus (Hacker Room)

Berlin, Germany

- Built generative-media systems with Gaussian splatting and diffusion models for spatiotemporal anchoring
- Implemented confidential-computing smart contracts in Rust
- Deployed product services and data infrastructure across Cloudflare Workers, D1, R2, Google Cloud Run and AWS EC2

### Senior AI Engineer

2024.07 - 2025.07

Vision Labs UG

Berlin & Hamburg, Germany

- Built workflows for prompt optimization, LLM evaluation and observability with DSPy and Langfuse
- Designed data and MLOps pipelines on Azure and Google Cloud with staging, automated tests and CI/CD

### Research Associate

2021.07 - 2024.07

Freie Universität Berlin

Berlin, Germany

- Scaled and optimized large neural-network simulations in C/C++ and Python
- Built retrieval-augmented generation (RAG) workflows for technical literature and evaluated AI-agent reliability
- Applied statistical and machine-learning analysis in Python; reviewed code and research manuscripts in an international team

Research Associate

2019.07 - 2020.06

Georg-August-Universität Göttingen • Göttingen, Germany

Neural networks, Partial Information Decomposition (PID), information theory and reproducible data workflows

Research Associate

2017.03 - 2019.07

Goethe-Universität Frankfurt • Frankfurt am Main, Germany • Simulation and statistical analysis of neural-network models

Visiting Scientist

2017.03 - 2019.03

Max Planck Institute for Dynamics and Self-Organization • Göttingen, Germany • Statistical methods for neural-network analysis

Research Associate

2015.02 - 2017.03

Forschungszentrum Jülich • Jülich, Germany • Neural-network simulations and reinforcement-learning experiments

Software Developer

2010.09 - 2012.03

Centre for Content Creation • Cyberjaya, Malaysia • Award-winning iPhone/iPad applications and digital products

EDUCATION

PhD Candidate (Dr. rer. nat.)

2020.04 - Present

Technische Universität Darmstadt • Darmstadt, Germany

- New mathematical PID analyses and information-theoretic representation analysis
- Engram’s domain-specific AI agent produces provenance-tracked neural-model candidates for closed loop neural network simulations, with bounded compilation and simulation checks, in reproducible computational-neuroscience workflows

Master’s by Research in Biosciences: Neuroscience

2012 - 2014

University College London • London, United Kingdom

Computational Neuroscience • Graduated with Distinction (Top 10%, CGPA 4.0/4.0)

Bachelor of Science (Honours) in Computer Science

2009 - 2012

Anglia Ruskin University • Cambridge, United Kingdom

Graduated with First Class Honours (Top 1%, CGPA 4.0/4.0)

SELECTED AWARDS & FUNDING

- EDTH / 2025  
Special Mention at the European Defense Tech Hackathon Berlin 2025; co-organized with the Bundeswehr Cyber Innovation Hub
- HUMAN BRAIN PROJECT  
Research funding

- NEAR HORIZON  
Confidential-computing grant
- UCL & SHEFFIELD  
Competitive academic scholarships
- IOS APPLICATIONS  
15 international product awards



SIGNATURE

Sepehr Mahmoudian

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PLACE Berlin, Germany

DATE 6 October 2026

PUBLIC ENGINEERING PORTFOLIO

GITHUB Open-source projects

[github.com/sepahead](https://github.com/sepahead)

HUGGING FACE Models & datasets

[huggingface.co/torusprime](https://huggingface.co/torusprime)



## SELECTED PROJECTS



### engram

★ 15 RESEARCH PROTOTYPE

#### PROVENANCE-CONSTRAINED NEURAL R&D

Domain-specific AI agent turns qualified evidence into neural-model candidates, with bounded compilation and simulation checks.

Neural Networks Knowledge Graphs Closed-loop Sim



### prisoma

★ 5 V0.9 PRERELEASE / ACTIVE R&D

#### AUDITABLE CLOSED-LOOP EXPERIMENTS

Simulation environment for closed-loop VLA and world-model experiments, with capture, replay and provenance.

Rust / Python VLA + World Models Embodied Sim



### manwe

★ 6 ALPHA

#### AIRSPACE PERCEPTION

Research workbench for vision, acoustics, multi-camera geometry and tracking; model exports must pass artifact-identity and numerical gates.

Python PyTorch Multimodal Perception



### melkor

★ 13 HARDENING

#### GAUSSIAN-SPLAT INTEGRITY

Inspects and deterministically converts 3D Gaussian splats, surfacing numeric hazards and field provenance before pipeline use.

C++ Scene Reconstruction Asset Validation



### galadriel

★ 8 RESEARCH PROTOTYPE

#### CROSS-SENSOR CONSISTENCY

Fail-closed monitor for sensor disagreement on one track, using per-channel innovation tests and signed correlation over producer-declared projections.

Rust NIS / CUSUM Signed Correlation



### pid-rs

★ 15 V0.9 SOURCE PRERELEASE

#### VERIFIABLE INFORMATION DECOMPOSITION

Measures how much information several signals share about an outcome, with built-in checks and provenance that keep results reproducible.

Rust / Python Information Theory Statistical ML



### NCP

★ 16 V0.8 RELEASE / 1.0 RC BLOCKED

#### NEURAL CONTROL PROTOCOL

Links neural simulators, AI controllers and robots through four typed planes with local fail-safes and provenance.

Rust Typed Control Planes Zenoh



### crebain

★ 22 RESEARCH PROTOTYPE

#### SPATIAL SITUATIONAL AWARENESS

Unifies Gaussian-splat scenes, native object detection and multimodal 3D tracking in one tactical view for simulated drone operations.

Rust Sensor Fusion Native Inference



### cobot-atlas

★ 8 +2,970 PUBLISHED + DOI

#### relief-atlas

AI-GENERATED SIMULATION DATASETS

2,023 robot-simulation and 10,079 disaster-relief GLB meshes with generation pipelines.

Python Hugging Face 3D / VLA



### cortexel

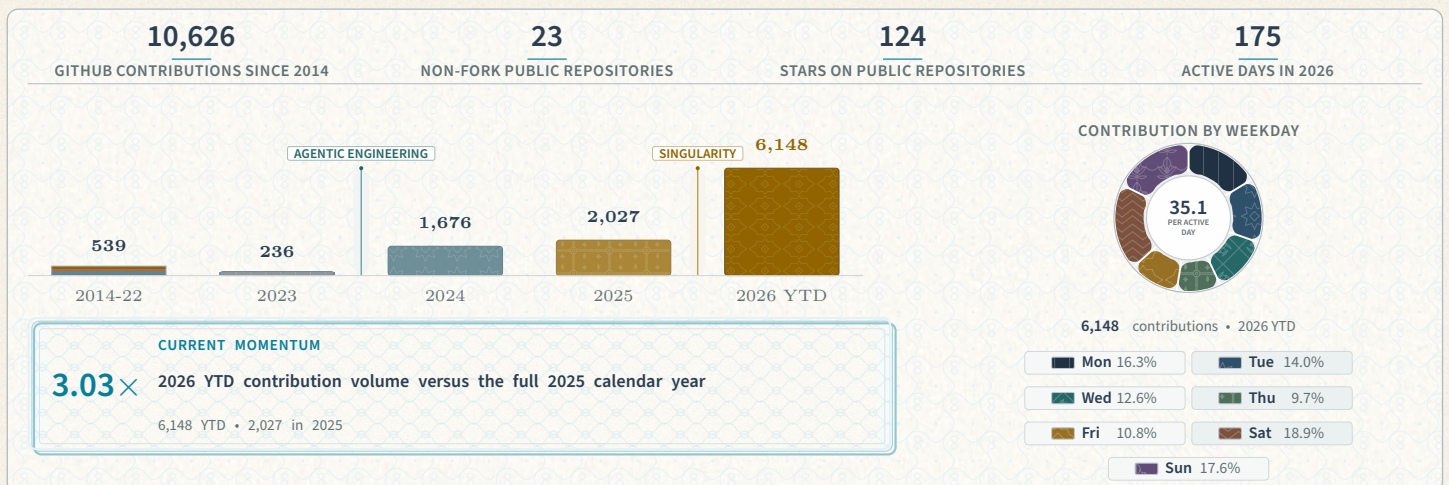
★ 5 0.10.0-DEV.0

#### AUDITABLE NEURAL FIGURES

Checks scientific figures against declared contracts before rendering, so charts provably match their data and stay reproducible.

Figure Contracts Deterministic SVG Provenance

## PORTFOLIO & ENGINEERING ACTIVITY



## TECHNICAL SKILLS

|                          |                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|
| • AGENTIC AI             | LLMs / VLMs   Agent Harnesses   AI Agents   Tool Use   Model Evaluation   Inference Pipelines |
| • MACHINE LEARNING       | Python   PyTorch   Agentic Vision   Multimodal AI   Vision-Language-Action / VLA Evaluation   |
| • KNOWLEDGE & RETRIEVAL  | Knowledge Graphs   Graph RAG   RAG   Embeddings   Vector Search   Reranking   Data Provenance |
| • SYSTEMS & MLOPS        | Rust   C++   FastAPI   ONNX   TensorRT   CUDA   Metal / Core ML   Docker                      |
| • CLOUD & INFRASTRUCTURE | GCP   AWS   Azure   Kubernetes   Terraform   GitHub Actions   Linux   Zenoh                   |